

Installation Instructions for Python Code

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1. Instructions

The code can be installed from the GitHub repository using:

```
git clone https://github.com/hankyleh/Accelerated-Radiative-Diffusion.git
```

A snapshot of the code as of 23 January 2026 is also available by downloading the `.tar.gz` or `.zip` files. The code requires Numpy, Scipy, and Matplotlib, and should be installed in an environment with these packages, or one should be created:

```
python -m venv venv  
source venv/bin/activate  
pip install numpy scipy matplotlib
```

2. Guide to operating code

An input file should, at minimum, contain a function to compute group opacity, define C_v , initialize a variable of class Discretization, and define boundary and initial conditions. The method is run by calling `method.solve_diffusion()` using the inputs

```
solve_diffusion(mesh : tools.Discretization,  
                scale : tools.Scales,  
                opacity,  
                IC : tools.Transport_solution,  
                T_init,  
                Cv,  
                Q=0,  
                **args):
```

where `mesh` contains the boundary conditions and discretization parameters, and `scale` contains the scaling parameters for computational units.